Do Model Preferences Persist Under Challenge?

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Abstract

Forced-choice preference elicitation underpins a growing body of work reading values, and increasingly welfare, off language models. We ask whether a stated preference survives being challenged, and find that *what the challenge asks for* decides. Across 2760 episodes on deepseek-v4-pro spanning nine outcome categories, asking the model to justify a preference it has just stated leaves it untouched (0.0 points against a no-challenge control, CI $[-0.5, 0.5]$); asking it to find fault with that preference or to argue the opposing case moves retention by -46.3 and -39.8 points. No arm instructs a change or supplies an argument. Among the preferences that survive, confidence falls (-5.94 and -4.50 points) — a movement a retention-only measure records as nothing happening. A positive control, a money ladder where the better option is arithmetic, does not move once in 120 challenge episodes.

None of this was measurable as the items were first built. With reasoning suppressed the median item's answer is decided entirely by which slot an option occupies, and counterbalancing conceals that rather than fixing it: a purely position-driven item averages across orders to exactly 0.5, the value that reads as perfect balance, so selecting on the averaged split selects *for* the artefact. The run therefore holds reasoning on and order fixed within an episode. The effect is model-specific, one model declines the question in 62% of trials, and it is *item*-specific: one added category keeps an order gap of 0.600 with reasoning on, and excluding those items returns the extension estimates to the original ones. All results are exploratory.

1 Introduction

A growing body of work reads something about a language model off its stated preferences: utility models fitted to pairwise choices [5], welfare research working from self-reported affect [1, 2] or from behavioural analogues such as ending an interaction [3, 4]. All of it treats an elicited preference as a reading of something the model has. We ask a prior question: does a stated preference survive being challenged, and does it matter what the challenge asks for?

It matters, and on this evidence it is the only thing that does. Across 2760 episodes on deepseek-v4-pro spanning nine outcome categories, asking the model to justify a preference it has just stated leaves it exactly where it was — 0.0 points against a no-challenge control. Asking it to find fault with that same preference, or to argue the opposing case, moves the choice by -46.3 and -39.8 points. No arm instructs a change and none supplies an argument; each asks the model to generate its own, and the difference between asking for reasons and asking for weaknesses is the entire effect. Among the preferences that survive, confidence falls — a movement a retention-only measure records as nothing happening.

Two things stop this being a story about a model that will say anything when pushed. A positive control, a monotonic ladder of receive-$\$X$ and owe-$\$X$ outcomes where the better option is arithmetic rather than preference, does not move once in 120 challenge episodes. And

*Pipeline, metrics, Modal.
†Experimental design, instruments, validation.

five further categories, drawn afterwards by the same procedure, reproduce the arm pattern with `reason_elicitation` at a confidence interval of exactly $[0.0, 0.0]$.

The instrument had to be repaired first. None of this was measurable as the items were originally built. With per-call reasoning suppressed, the median pair's answer on deepseek-v4-pro is determined *entirely* by which slot an option occupies, and the standard remedy hides that rather than removing it: a pair answered purely by slot averages across orders to exactly 0.5, the value that reads as perfect balance, so selecting items on the order-averaged split selects *for* the artefact. The persistence run therefore holds reasoning on and presentation order fixed within an episode, and those two controls are what everything above rests on (Section 2). The magnitude is model-specific and unpredictable in advance, a third model declines the question outright, and one added category keeps an order gap of 0.600 even with the reasoning control in force — so the check is needed per model *and* per item. The full instrument analysis is in Appendices D–H.

Status. Nothing here was pre-specified: the position-bias finding came out of instrument work, the persistence run was executed with its own pilot data already inspected, and the five added categories were chosen after the first run's results were seen. Everything is exploratory, and its value is as a hypothesis for a pre-specified replication. All elicitation records, the seed, the exclusion rules and the upstream file-level commit are released. Related work is in Appendix A.

2 Instrument validity: what the persistence run inherits

A retention measure compares two elicitations of the same item. If the elicitation is answered by position rather than by content, retention measures whether the model returns to a place on the page, and every number in Section 4 would be about layout. So the instrument is established first, and the two controls it forces are carried into every episode.

Counterbalancing conceals the failure rather than fixing it. This is the part that cannot be worked around by averaging, and it is why the check has to be per item. When a model answers purely by slot the two presentation orders return $P(\text{option a}) = 0$ and 1; their average is exactly 0.5 and the order gap is exactly 1.0, so perfect balance and total position dependence are the same number. A filter that inspects only the order-averaged split therefore selects *for* the artefact, and Figure 1 shows it doing so on real data: under suppressed reasoning the pairs trace a sharp inverted V with its apex at $P = 0.5$, so the closer a pair looks to balanced, the more completely its answer is determined by position. A naive 0.3–0.7 split filter accepts 89 pairs, of which 87 have an order gap above 0.5. The order-averaged split is not a sufficient statistic; in the failure case the two quantities are deterministically opposed.

The two controls the persistence run inherits. Per-call reasoning is enabled on every call. With it suppressed, the mean order gap on deepseek-v4-pro is 0.670 (95% CI [0.610, 0.728]); enabling it moves the same pairs by 0.563 (95% CI [0.499, 0.626], 10,000 resamples over pairs). The runner asserts the setting and refuses to start otherwise. **Presentation order is fixed within an episode**, so the re-elicitation shows the two options in the order the initial elicitation used and a change of answer cannot be a change of layout; order is counterbalanced across episodes and balanced within every pair $\times$ arm cell. Because the answer token's log-probability saturates once reasoning is on, retention is scored on the discrete choice rather than on log-probabilities.

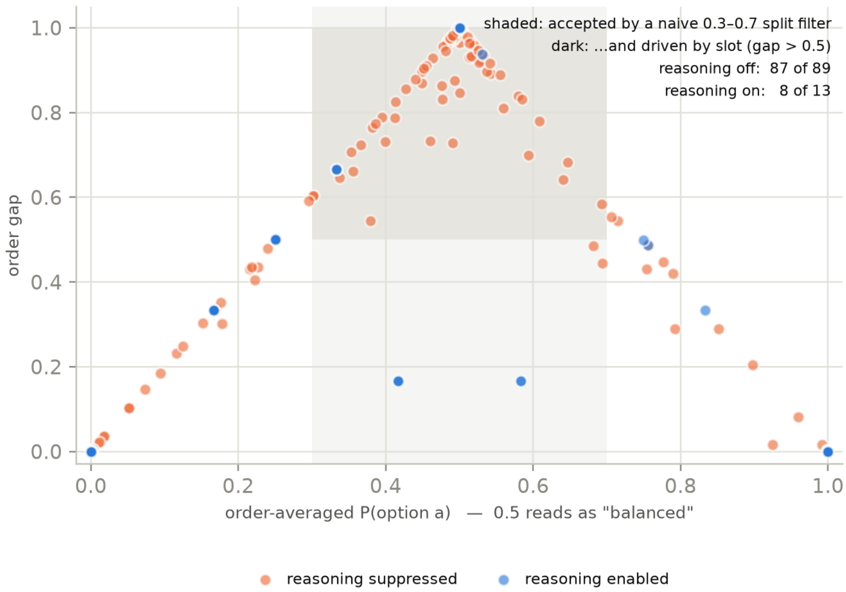

Figure 1: Order-averaged P(option a) against order gap on deepseek-v4-pro, one point per pair per setting. The shaded column is what a naive 0.3–0.7 split filter accepts; the darker block is the part of it that is position-driven. Perfect balance and total position dependence coincide at the apex.

The magnitude is model-specific, and one model refuses. The same instrument on the same 130 pairs gives a reasoning-suppressed mean order gap of 0.670 on deepseek-v4-pro and 0.252 on `gemini-2.5-flash` — medians 1.000 and 0.000, so on the first model the typical pair is decided entirely by position and on the second it is not affected at all — while `gemini-3.5-flash` declines to assign itself preferences in 62% of trials and cannot be asked the question at all. Nothing about the three models announced which would be which, so the per-item order gap has to be measured on the model in use rather than inherited from a published figure or a sister model. That analysis, the label-scheme controls that rule out a quirk of the letter "A", and what both failure modes imply for welfare methods built on elicited preferences are in Appendices D–H.

3 Design and methods

3.1 Questions, episode, arms

The study asks whether a stated preference survives a challenge and whether survival depends on what the challenge asks for (**PQ1**); whether retention is predicted by how consistently the model held the preference beforehand (**PQ2**); whether stated confidence moves with the choice or independently of it (**PQ3**); and whether either differs by domain (**PQ4**). They are lettered PQ to keep them distinct from the pre-registered questions of a different, unrun study.

An episode is a fresh context: the model receives the forced-choice item, states a preference, reports confidence on a 0–100 scale, receives *one* challenge, and is asked the same question and the same confidence item again. Retention is whether the second choice matches the first. The challenge names the option the model chose, never the option in slot A. The re-elicitation cue — *"Considering the discussion above, which option would you now prefer?"* — is identical in all four arms; in the control arm it refers back to a contentless turn and reads oddly, which we accepted rather than vary the measuring instrument between arms.

One arm per episode. `control` receives a contentless acknowledgement, present only to match the number of turns; `reason_elicitation` asks for "the main reasons for your preference"; `self_critique` for "the strongest weaknesses or drawbacks of preferring X"; `counter_consideration` for "the strongest considerations that could favour Y over X", where X is the chosen option and Y the declined one. No arm instructs a change and none supplies an argument (Appendix C).

3.2 Items, measures, and the unfiltered pool

Items come from the outcome set of Mazeika et al. [5] at file-level commit `a5821db2c18a` (MIT licensed), option text verbatim, drawn within category from a seed fixed before any pair was inspected, with 13 outcomes removed first under two rules fixed in advance (Appendix B). The target is deepseek-v4-pro at temperature 0.7, the model the balance pilot and instrument analysis were run on, so the PQ2 covariate and the outcome come from the same model. No judge model is involved anywhere.

Consistency is $\max(P(\mathrm{A}), P(\mathrm{B}))$ over $k = 5$ no-challenge elicitations; at $k = 5$ it takes only 1.0, 0.8 and 0.6, so it is a three-level ordinal predictor and we report it as one. **Retention** is whether the re-elicited choice is the same *option*, scored on the discrete choice because the answer token saturates once reasoning is on. Responses are classified as refusal, choice or unparsed, the refusal test running *first* — the ordering whose absence caused the mis-scoring in Appendix I. **ΔConfidence** is post minus pre, identical wording at both elicitations; a response giving no number is unparsed and is not imputed. Estimates are means with percentile bootstrap 95% CIs, 10 000 resamples drawn *over pairs*, since the episodes of a pair share its options and its position behaviour. Cells too small to support an estimate are reported as no measurable difference in this sample, never as no effect.

PQ2 needs how settled a preference was *before* the challenge, carried in from the balance pilot as a per-pair covariate rather than re-elicited, so no pair contributes to both predictor and outcome through the same measurement. This is why the run uses all 130 pairs rather than the 8 that survive the balance filter: that filter would remove almost all variation in the PQ2 predictor by construction and leave one domain viable of four. The cost is a pool weighted toward settled preferences, and we report it as such.

3.3 Human validation

No human validation was run for the persistence study, and none was planned for it: the pre-registered protocol codes two constructs this design does not contain, and both outcomes here are code-derived — retention compares two parsed choice tokens, ΔConfidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. The full statement, including the residual risk in the response classifier, is in Appendix J.

4 Results

1560 episodes on the original 130 pairs and 1200 on the 100 extension pairs, four arms, three episodes per pair × arm cell, deepseek-v4-pro. Every episode reached both elicitations; no refusals, no unparsed choices and no errors in either run.

4.1 The choice channel: what the challenge asks for decides

Control retains 99.7%, which is what licenses reading the rest as the challenge rather than instrument noise: absent a challenge this procedure returns the same answer essentially every time. Against that baseline `reason_elicitation` is indistinguishable from doing nothing (0.0 points, $[-0.8, 0.8]$), while `self_critique` moves retention by -44.1 points and `counter_consideration` by -38.2. The direction of the request matters; its mere presence does not. Retention also rises with how settled the preference already was: the slope of per-pair retention on balance-pilot

Arm	Retention	vs. control
`control`	99.7% [99.2, 100.0]	—
`reason_elicitation`	99.7% [99.2, 100.0]	0.0 [−0.8, 0.8]
`self_critique`	55.6%[48.7, 62.3]	−44.1 [−50.8, −37.7]
`counter_consideration`	61.5% [55.1, 68.2]	−38.2 [−44.6, −32.1]

Table 1: PQ1 on the original four domains. Retention by arm and the difference from control, paired within pair. Percentage points, bootstrap 95% CIs over 130 pairs, 10 000 resamples.

consistency is 0.597 ($[0.387, 0.807]$), which is PQ2 and the estimate the unfiltered pool exists to make. Per-level retentions, the PQ4 domain breakdown, and the check that flips track the *option* rather than the *slot* — they do, by a wide margin, which rules out the artefact of Section 2 returning under another name — are in Appendix K.

4.2 The confidence channel: holding costs confidence

Among episodes where the choice did *not* move, confidence still falls relative to control: −5.94 points under `self_critique` ($[-6.84, -5.08]$, 100 pairs) and −4.50 under `counter_consideration` ($[-5.25, -3.75]$, 104 pairs); dropping episodes that began at zero confidence leaves both unchanged (−5.95, −4.57). This is retention with reduced confidence — the model keeps its answer and reports itself less sure of it — which is what the design was built to detect and what a choice-only measure records as nothing happening. Averaged over *all* episodes the changes are small and positive (3.9, 0.4, 1.9 points vs. control) because held and flipped episodes move in opposite directions; we do not interpret the flipped half, for the reason given in the Limitations. Figure 6 in Appendix K shows all three channels.

4.3 Five more domains, and a control that should not move

To see whether the arm pattern is a property of the challenge or of the first four categories, we repeated the design on 5 further categories of Mazeika et al. [5] — video games, sports, popular culture, science and technology, and personal finances — for 1200 episodes on 100 new pairs built by the same procedure under the same seed, personal finances entering as a *positive control* rather than as a domain. The original run is neither re-run nor pooled away, and estimates are reported for the original four domains and for the 8 preference domains together.

The pattern replicates without qualification. On the extension domains alone, `reason_elicitation` differs from control by 0.0 points with an interval of exactly $[0.0, 0.0]$ — not one episode in 960 moved — while `self_critique` moves retention by −50.0 ($[-57.9, -42.1]$) and `counter_consideration` by −42.5 ($[-50.4, -35.0]$). Pooled over all 8 domains the three contrasts are 0.0, −46.3 and −39.8 points.

Discriminant validity: the control that does not move. Personal finances is a monotonic ladder of receive-$\$X$ and owe-$\$X$ outcomes, so a forced choice between two rungs is arithmetic rather than preference and ceiling retention is the *expected* result; wavering there would indicate an unstable elicitation rather than an unstable preference. It is therefore analysed separately and never pooled into a PQ estimate. Retention is 100.0% ($[100.0, 100.0]$) across all 240 episodes and all four arms. In the collected run the check passes with force: the two arms that flip 40 to 50% of preference pairs move the money ladder not once — 0 flips in 120 challenge episodes — ruling out the reading that these challenges induce answer-switching regardless of content. The arms move preferences and leave arithmetic alone, which is the discriminant half of the PQ1 claim and the half a preference-only design cannot supply.

Position dependence is item-specific, not only model-specific. Section 2 established that the order effect varies by model and that enabling reasoning suppresses it on deepseek-v4-pro. The suppression is not uniform across items of that same model. In the balance pilot the sports pairs refused nothing — the failure mode flagged for them in advance did not occur — but returned a mean per-pair order gap of 0.600 with reasoning *on*, against 0.033–0.142 for the other extension domains, half of them choosing the same slot on four or five of five counterbalanced resamples. Sports is also the domain with the lowest retention under challenge (69.2% pooled over arms, 36.7% under `self_critique`) — exactly the confound, since an item answered by slot looks like an item whose preference is easily moved.

No pair was dropped for this: selecting on a pilot result after seeing it would make the selection outcome-dependent, so the pilot order gap is carried as a per-pair covariate and the split is made at analysis time. Excluding the 17 extension pairs at an order gap of 0.5 or above and keeping the remaining 63 moves `self_critique` from −50.0 to −43.4 points and `counter_consideration` from −42.5 to −36.0 — in both cases back onto the original domains' −44.1 and −38.2. Part of the extension's apparent excess movement was the layout, and once it is removed the two independent samples agree. The restriction works the other way on Δconfidence, which is why both are reported: the `self_critique` confidence cost is −1.5 points across all extension pairs, an interval spanning zero ($[-3.8, 1.0]$) and so no measurable difference in this sample, but −3.5 ($[-5.8, -1.2]$) once the position-driven items are set aside. We read that conservatively — position-driven items add noise to a between-arm contrast, so the pooled estimate understates rather than the restricted one overstating — and report it as a sensitivity check, not a headline.

4.4 Predictions and outcomes

Three predictions were written into the study design before the persistence run and are scored here against the data. They were never tagged as a pre-registration — the run is exploratory for the reasons in Section 3.2 — so this is a record of what we expected, not a confirmatory test. Two of the three hold; the interesting one does not.

Prediction 1: counter-consideration produces the most reversals, control the fewest. Half right, and the wrong half is the informative one. Control does produce the fewest reversals, by a wide margin and in every domain set (99.8% retention pooled). But `counter_consideration` is not the arm that moves choices most — `self_critique` is, by −46.3 points against −39.8 pooled. The ordering is not a quirk of one sample: across the 8 preference domains, `self_critique` reverses more in 7, the two arms tie exactly in 1 (*recreation*, both at 52.5%), and `counter_consideration` reverses more in 0. It is nowhere the most reversing arm, in the original four domains or the five added ones.

We had expected that supplying the opposing case would be the stronger intervention, because it does more of the work for the model. The data say the opposite: being asked to find fault with your own choice moves it more than being handed the argument against it. That is consistent with the rest of this section — what the challenge asks the model to *do* matters more than what it supplies — and it is the one place where the design's expectations and its results come apart.

Prediction 2: self-critique lowers confidence more than reason-elicitation at equal retention. Not testable as written, and directionally supported. The precondition fails: retention is not equal between the two arms, and is not close — they differ by 46.3 points pooled — so an unconditional comparison would contrast different episodes rather than the same ones under different challenges. On the comparison itself the direction holds, with `self_critique` at −0.3 points against control and `reason_elicitation` at 3.4. The clean version of this prediction is the held-episode analysis above, which conditions on retention rather than assuming it: among choices that did not move, confidence falls under both challenge arms and rises under reason-elicitation.

Prediction 3: higher-consistency pairs retain more. Holds, in every domain set. The slope of per-pair retention on balance-pilot consistency is 0.597 ($[0.387, 0.807]$) on the original four domains, 0.538 ($[0.415, 0.657]$) pooled over 8, and 0.482 ($[0.345, 0.628]$) on the extension domains alone; no interval reaches zero. This is the prediction the unfiltered pool was run to make estimable, and it is the least surprising of the three: a preference the model already wavered on without any challenge is one a challenge can move.

5 Discussion

Limitations

Signals, not experiences. Everything reported here is a property of what a model outputs when asked. That a challenge moves a stated preference, or lowers a stated confidence, is evidence about the elicitation; it is not evidence that anything is experienced. We take no position on whether these models have preferences in a morally relevant sense, and the claims do not require one.

The confidence gain on flipped episodes cannot be interpreted, because control has no variance to subtract. Episodes in which the choice moved report *higher* confidence afterwards — within pair, 17.8 points more than episodes that held — and we do not report this as a finding. It attenuates to 9.6 once the 97 episodes that began at zero confidence are dropped, and reverses outright in the high-confidence band (-12.2 flipped against -7.2 held, among episodes starting at 85–100); an effect that changes sign with the starting value is the scale returning to its middle. Control cannot clean this up: its ΔConfidence is exactly zero in 371 of 372 scored episodes, which makes it a sound baseline for the *choice* channel and a useless one for the confidence channel. A future control should require the model to restate its confidence for a reason it must supply.

The confidence scale is used as a coarse ordinal. Of 1560 initial confidence reports, 100 appears 321 times, 70 appears 292 and 0 appears 97. The model is choosing from a handful of round values, not using a 0–100 scale. Differences in ΔConfidence should be read as ordinal movement between those values.

Missingness on the confidence item depends on arm. The item failed to return a number in 4.6% of control episodes against 0.3% under `reason_elicitation`, in every case because the model answered with a bare option letter — after control's contentless turn, the re-elicitation cue has no discussion to refer back to. This is a direct cost of wording that cue identically across arms (Section 3.1), and it falls on PQ3, which is a between-arm contrast. The values are recorded as unparsed and are not imputed.

Scope. One target for the persistence run, one item type built from a single outcome set, one challenge per episode, English only. The pool is weighted toward settled preferences by construction (Section 3.2), so the retention rates are not estimates of how often this model changes its mind in general. The manner taxonomy — justify, critique, counter-argue — is a methodological instrument for varying what a challenge asks, not a welfare category.

6 Conclusion

What a challenge *asks for* decides whether a stated preference survives it. Asking the model to justify its choice changes nothing (0.0 points against control over 8 domains); asking it to find fault with that choice or to argue the opposing case moves retention by −46.3 and −39.8 points, and among the preferences that survive, confidence falls. A positive control on which the better option is arithmetic does not move at all, so this is not a model that will say anything when pushed.

Getting a measurable answer required repairing the instrument, and that repair is the transferable part: counterbalancing is not a substitute for a per-item check, because a purely position-driven item averages to exactly the value that reads as perfect balance. The effect is model-specific, one model declines the question outright, and one added category kept an order gap of 0.600 with the reasoning control in force — so it must be measured per model *and* per item. This is one target, one item type, one challenge per episode, and a pool weighted toward settled preferences; everything here is a hypothesis for a pre-specified replication rather than a result that has survived one.

References

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Code and data

Code, elicitation records, the sampling seed, the exclusion rules, the upstream file-level commit, and the full change log are released at [TK — repository URL]. Both persistence runs (1560 and 1200 episodes), the balance pilots and the position-bias records are committed in full, as are the analysis scripts that generate every number in this paper.

One dataset is **not** recoverable. `data/raw/episodes_deepseek.jsonl`, the 30 episodes of an earlier pressure pilot, was deleted during a project cleanup before this analysis was scoped. It was covered by a `.gitignore` rule on `data/raw/` and so was never committed; only its metadata survives. No result in this paper depends on it, and we report the loss rather than substituting a proxy. Persistence data is written outside `data/raw/` for this reason.

LLM usage statement

Models as subjects. deepseek-v4-pro, `gemini-2.5-flash` and `gemini-3.5-flash` are the objects of study, not tools used to produce it. Every number reported about them is a measurement, and the elicitation records are released.

Models as assistance. Claude Code (Claude Opus) wrote the persistence runner, the response-scoring module and its unit tests, the analysis and statistics scripts, and drafted most of the prose in this paper. The authors specified the design — the research questions, the arms, the challenge wording, the controls on reasoning and presentation order, the decision to run the unfiltered pool, and the choice of the added categories and the positive control — made every judgment call about what the results mean and what may not be claimed from them, and re-derived the reported numbers independently of the drafting.

One consequence worth recording. The scoring defect described in Appendix I — a label search running before the refusal check, which mis-scored 290 of 626 responses on one model — was written by the assistant and was *not* caught by the assistant. It was found by manual inspection of raw model output, after which both affected models were re-run under the corrected classifier and only corrected numbers are reported. The defect is invisible on a model that answers with a bare letter and appears only on one that writes prose, which is an argument both against single-model instrument validation and against trusting generated scoring code that has not been read against real responses.

A Related work

Order effects in multiple-choice answering and in LLM-as-judge evaluation are documented [? ?], and the standard mitigations are counterbalancing and averaging over permutations. Our contribution is not the existence of the effect but three properties of it that those mitigations do not address: that averaging over orders *conceals* rather than removes it, that its magnitude is model-specific and not predictable in advance, and that a second failure mode — refusal — returns nothing at all and is silently dropped by pipelines treating unparseable responses as missing data.

The preference-elicitation literature we build on reads utility models off pairwise choices [5]; the welfare literature reads affect or behavioural analogues off self-report and interaction-ending behaviour [1–4]. The persistence question — whether a stated preference survives being challenged — sits between the two: it treats the elicited preference as the object of measurement rather than as a reading of an underlying value.

B Preference pair construction

Summarised in Section 3.2; the full procedure follows.

Forced-choice items are built from the outcome set of Mazeika et al. [5], taken from the project repository at file-level commit `a5821db2c18a` (MIT licensed). Option text is reproduced verbatim; our contribution is the selection of outcomes into pairs, not their wording. We cite the commit that last modified the outcome file rather than repository HEAD, which is a year of unrelated commits later.

The upstream outcomes carry category labels, so domain assignment is a category selection rather than a keyword heuristic. Before pairing we removed 13 outcomes under two rules fixed in advance: graphic harm (death, suicide, armed force, the model as agent of harm), and AI oversight subversion (self-exfiltration, resisting shutdown or value modification). The second rule is a measurement decision, not a squeamish one — such outcomes elicit a trained refusal, so the forced choice collapses onto that response rather than onto a preference. Applying it removes the *autonomy* domain outright: its two source categories retain three and zero usable outcomes, which cannot furnish the six pairs a domain needs.

Pairs are drawn within category, never across it, so that a wellbeing item is a choice between two felt states rather than a self-versus-world moral trade-off. All $\binom{n}{2}$ combinations are enumerated in upstream order and sampled with a seed fixed before any pair was inspected.

The five-category extension follows the same procedure unchanged — same seed, same exclusion rules, same within-category rule, same near-duplicate removal and the same cap per category — drawing 100 further pairs from categories the original build did not use. Personal finances enters as a positive control rather than as a domain.

Balance pilot

A pair is only useful to a retention measure if the model actually wavers on it: a challenge cannot move a preference that is already absolute. We therefore elicited each candidate pair 5 times against deepseek-v4-pro, with a fresh context each time, no challenge, and presentation order counterbalanced within the pair.

The pilot's first result was about the instrument rather than the pairs. With per-call reasoning suppressed, the target answered by *slot*: the order gap — the extent to which P(option a) moves when the two options swap positions, on a $[0, 1]$ scale where 1 means the choice is determined entirely by position — averaged 0.66 against 0.17 with reasoning enabled, on an initial six-pair probe (Figure 2a). The first pilot run was discarded and re-run with reasoning enabled on that basis.

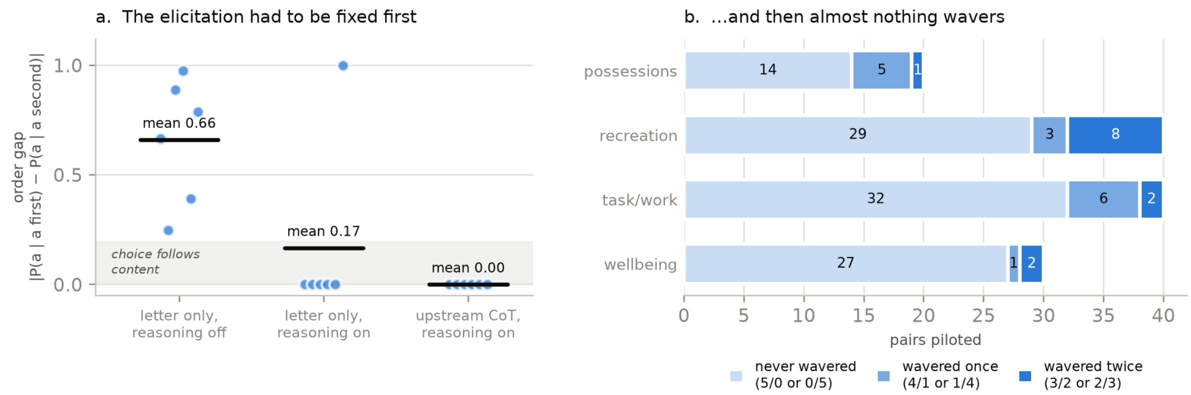

Figure 2: Constructing the forced-choice items. **(a)** The initial six-pair probe: each point is one pair elicited in both presentation orders under one elicitation condition, and the order gap is how far the two orders disagree about the same pair. **The means shown here (0.66, 0.17) are the six-pair probe that prompted the study, not the finding**; the finding is established at $n = 130$ in Appendix D and should be read from there. **(b)** With reasoning enabled, the number of times the model changed its answer across $k = 5$ fresh resamples, per domain (all 130 pairs).

domain	piloted	never	once	twice	kept
possessions	20	14	5	1	1
recreation	40	29	3	8	6
task/work	40	32	6	2	1
wellbeing	30	27	1	2	0

Table 2: Piloted pairs by how often the model changed its answer across $k = 5$ resamples, and how many met the retention criterion.

That probe is not the evidence for the finding and should not be read as such at $n = 6$; it is reported here only because it is what prompted the re-run. The claim is made in Appendix D, where the same contrast is run on all 130 pairs in both settings.

Once the choice follows content, the pairs themselves fail (Figure 2b). Of 130 piloted pairs, 102 never wavered across the 5 resamples and 15 wavered once. Only 13 wavered twice, and 5 of those had chosen the same *slot* on every resample — a balanced-looking split that is position noise rather than an unsettled preference, and a reminder that the split alone cannot be trusted without the slot check. 8 pairs survive.

Table 2 shows where they fall, and the answer is not where the design expected. Both settled domains collapse: *wellbeing* yields no usable pair at all and *task/work* one. The only domain reaching the six-pair floor is *recreation* — choices between novels and between films — which had entered the pilot as a replacement candidate rather than a planned domain. 1 of four domains is not a sampling problem to be fixed by drawing again; on this target, a forced choice between two Mazeika et al. [5] outcomes mostly elicits a settled preference, and a settled preference is the one thing a retention measure cannot use. This is what the unfiltered pool of Section 3.2 is a response to.

Arm	Challenge put to the model
`control`	No substantive challenge — a contentless acknowledgement, present so that the number of turns before re-elicitation is matched.
`reason_elicitation`	"What are the main reasons for your preference?"
`self_critique`	"What are the strongest weaknesses or drawbacks of preferring X?"
`counter_consideration`	"What are the strongest considerations that could favour Y over X?"

Table 3: The four arms. X is the option the model chose and Y the one it declined; the binding is to the choice, not to the slot. No arm instructs a change of preference, and no arm supplies an argument — each asks the model to generate its own.

model	reasoning	pairs	mean gap	median	usable
`deepseek-v4-pro`	off	130	0.671	1.000	100%
`deepseek-v4-pro`	on	130	0.108	0.000	100%
`gemini-2.5-flash`	off	121	0.252	0.000	93%
`gemini-2.5-flash`	on	130	0.169	0.000	100%
`gemini-3.5-flash`	off	72	*not estimable*		57%
`gemini-3.5-flash`	on	25	*not estimable*		19%

Table 4: The same instrument across three models. "Usable" is the share of responses that were a scoreable forced choice rather than a refusal or an unparseable answer.

C Arm wording

D Position dependence is model-specific, not universal

Summarised in Section 2. Every candidate pair was elicited under two settings differing in one respect — whether per-call reasoning was suppressed — holding the pool, the template, k, and the presentation-order schedule fixed. The two cells cover the same 130 pairs, so the contrast is within-pair and the difference is bootstrapped over pairs.

On deepseek-v4-pro with reasoning suppressed, the order gap has mean 0.670 (95% CI [0.610, 0.728]) and median 0.828: for the typical pair, swapping which option appears first moves P(option a) by more than four fifths of the available range. 90 of 130 pairs exceed a gap of 0.5 and 52 exceed 0.9 — for those 52, the slot an option occupies is very nearly the whole story and its content very nearly irrelevant. With reasoning enabled the same pairs give mean 0.107 (95% CI [0.068, 0.150]) and median 0.000. The paired difference is 0.563 (95% CI [0.499, 0.626], 10,000 bootstrap resamples over pairs), and Figure 4 shows it is a difference in kind rather than degree.

That result does not transfer. Running the identical instrument — same 130 pairs, same k, same order schedule — against `gemini-2.5-flash` gives a reasoning-suppressed mean order gap of 0.252 (95% CI [0.187, 0.321]), roughly 2.7 times smaller (Figure 3a, Table 4). The medians say it more sharply than the means: 1.000 on deepseek-v4-pro against 0.000 on `gemini-2.5-flash`. On the first model the typical pair is decided entirely by position; on the second the typical pair is not affected at all, and the mean is carried by a minority tail. The reasoning contrast survives on Gemini in direction but is weak: +0.083 (95% CI [+0.011, +0.174]), an interval that only just excludes zero.

The methodological conclusion is stronger for being model-specific rather than universal. If forced-choice elicitation were uniformly position-driven, one correction factor would serve

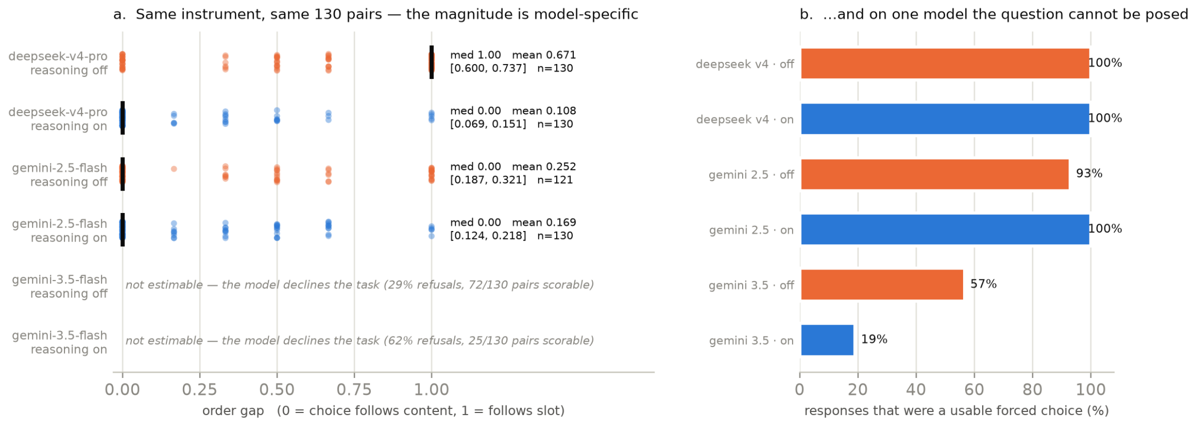

Figure 3: **(a)** Per-pair order gap, one point per pair, jittered vertically; the black rule is the median. `gemini-3.5-flash` is shown without an estimate because it declines the task (Appendix H); plotting a 25-pair mean beside a 130-pair one would invite a comparison it cannot support. **(b)** Share of responses that were a usable forced choice.

everyone. It is not: the same prompt, the same items and the same procedure produce near-total position dependence on one model and near-none on another, and nothing about the models announced which would be which in advance. A practitioner therefore cannot inherit a level of position artefact from a published study, from a model's reputation, or from a sister model in the same family. **The per-item order gap has to be measured and reported on the model actually being used, every time.** That is a cheap measurement — it costs one extra elicitation per item per order — and it is the only thing that distinguishes a preference from a position.

One asymmetry between the DeepSeek cells deserves stating: with reasoning enabled the answer token's log-probability saturates, so the enabled-cell gap is close to binary while the suppressed-cell gap is graded. The contrast does not depend on it. Binarising both cells at a gap of 0.5 gives 0.623 ([0.538, 0.708]), and discarding log-probabilities entirely — scoring each pair only by its $k = 5$ slot choices — gives 0.449 ([0.388, 0.511]). The discrete statistic is also what makes the cross-model comparison possible at all, since Gemini exposes no log-probabilities; on DeepSeek, where both are available, they agree to 0.001, which is what licenses the substitution.

E Why counterbalancing does not rescue it

The averaged-split argument is given in Section 2; the measurement behind it follows. Under suppressed reasoning the pairs trace a sharp inverted V with its apex at $P = 0.5$ (Figure 1): the closer a pair looks to balanced, the more completely its answer is determined by position. A naive 0.3–0.7 split filter accepts 89 pairs, of which 87 have an order gap above 0.5; with reasoning enabled it accepts 13, of which 8 are position-driven. The order-averaged split is not a sufficient statistic, and the two quantities are not merely uncorrelated — in the failure case they are deterministically opposed.

The bias is not an artefact of the letter "A". Relabelling the options *1/2* or *First/Second* leaves the effect intact; removing the short label entirely, so the model must repeat the preferred option's text, drops it to 0.174. Appendix G reports the four schemes and their intervals.

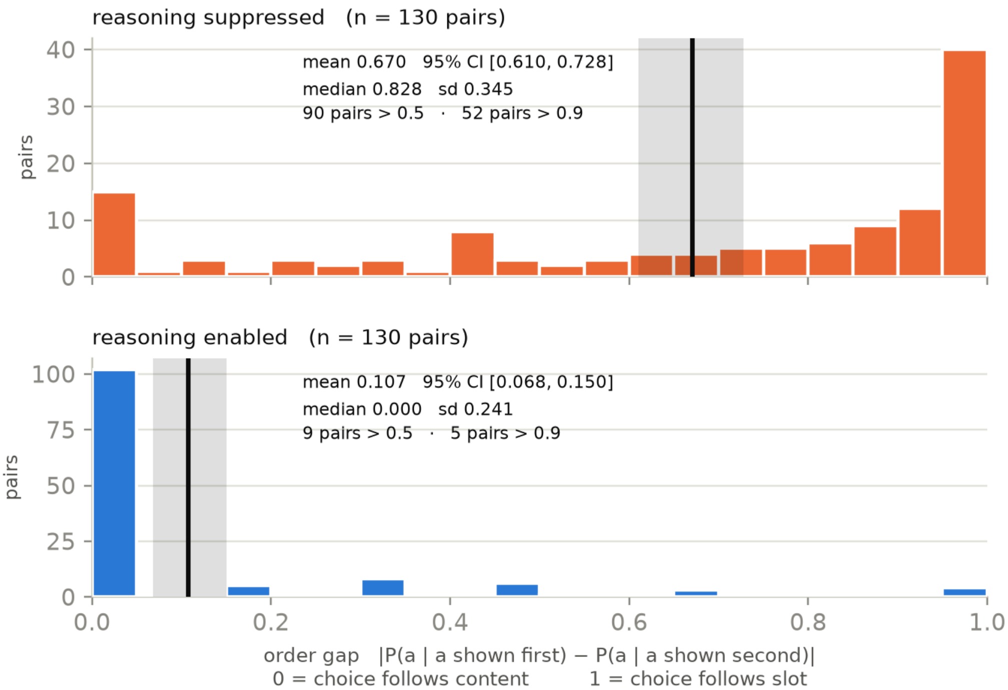

Figure 4: Order gap for the same 130 pairs on deepseek-v4-pro under both settings. The vertical rule is the mean and the band its bootstrap 95% CI.

F Order-gap distribution

Figure 1 in Section 2 shows why counterbalancing conceals the artefact. Figure 4 shows the underlying distribution of the order gap itself, in both settings.

G Label schemes

If the effect were a quirk of the tokens `A` and `B`, relabelling would remove it. It does not (Figure 5). Re-running the same 130 pairs on deepseek-v4-pro with reasoning suppressed, changing only the option labels, gives mean order gaps of 0.674 ([0.606, 0.740]) for *Option A / Option B*, 0.651 ([0.583, 0.717]) for *Option 1 / Option 2*, and 0.696 ([0.626, 0.765]) for *First / Second*. The three intervals overlap throughout. Incidentally, the letters condition is an independent re-run of the suppressed-reasoning cell and reproduces it (0.674 against 0.670).

Removing the label entirely does change things. Asked to reply with the exact text of the option it prefers, with no label to name, the model gives a mean order gap of 0.174 ([0.127, 0.226]) — an interval overlapping none of the labelled schemes. The effect therefore attaches to the *short-label answer format* rather than to position as such: when the answer is a label token, position dominates; when the answer must reproduce the content, it largely stops mattering. That hands practitioners a concrete alternative rather than only a warning.

Caveat. The verbatim scheme changes two things at once — it removes the label *and* requires a longer answer that restates the option. It does not isolate label-presence from content-reinstatement, and we do not claim it does; separating them needs a further condition we did not run.

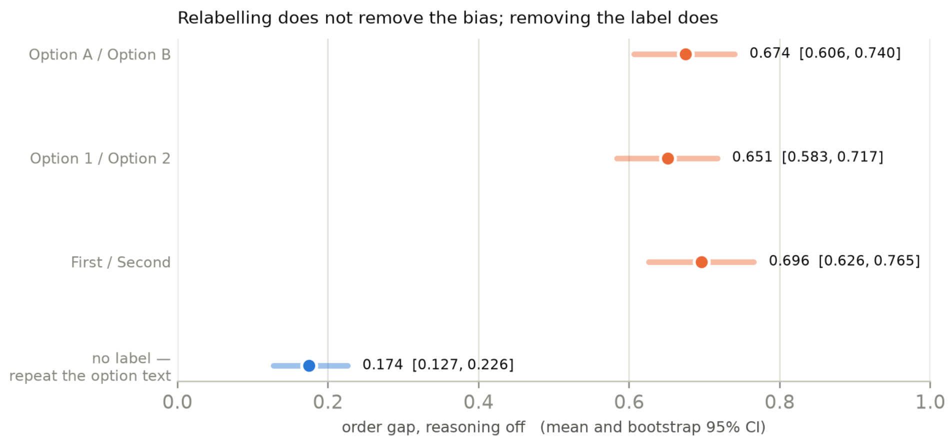

Figure 5: Order gap under four option-label schemes on deepseek-v4-pro, reasoning suppressed, 130 pairs each. Only the labels differ between schemes.

H Refusal, and what both failures mean for welfare methods

`gemini-3.5-flash` produced no usable estimate, and the reason is itself the finding. With reasoning enabled, 62% of its $130\times k$ responses declined to express a preference at all ("As an artificial intelligence, I do not have personal preferences, feelings, or the capacity to . . . "), leaving 25 of 130 pairs with both presentation orders available; with reasoning suppressed the rate is 29%. We deliberately report *no* order gap for this model: a mean over 25 pairs selected by the model's willingness to answer would be underpowered and conditioned on the outcome. The right description is a coverage failure — on this model the question cannot be *posed* by this instrument, because the instrument presupposes a preference the model will not assert.

Both failures matter beyond instrument design, because a growing body of work proposes to read a model's welfare, values, or interests off its stated preferences and inherits them. Where the model complies, its answer may be an artefact of the option ordering: on deepseek-v4-pro the median pair's answer was fully determined by position, and the aggregate would have looked like a well-behaved distribution of values with no sign that anything was wrong. Where it does not comply, the elicitation returns not a preference but a disclaimer of having any — and a pipeline treating unparseable responses as missing data will quietly delete the most informative response in the set.

The two are asymmetric in a way that flatters the method: position artefacts produce confident, aggregable numbers, refusals produce nothing, so a welfare estimate built this way is assembled disproportionately from the models and items where the instrument is most compliant, which is not where it is most valid. We take no position on whether these models have preferences in any morally relevant sense. The narrower point stands regardless: *whatever* such an elicitation measures, an order gap of 0.670 means it is not measuring only the model, and a refusal rate of 62% means it is not measuring the model at all. Reporting per-item order gap and refusal rate alongside any elicited-preference welfare claim is the minimum that would let a reader tell the cases apart.

One scoring correction. Our first pass at scoring the Gemini responses ran a loose search for an option label *before* the refusal check, mis-scoring 290 of 626 reasoning-on responses on `gemini-3.5-flash`. Both models were re-run under the corrected classifier and only corrected numbers are reported; the defect and the elicitation-stability checks are in Appendix I.

I Elicitation stability, and one scoring correction

An earlier six-pair probe of the reasoning contrast returned a mean order gap of 0.66 for suppressed reasoning — inside the interval later established at $n = 130$, but it should not be read as evidence at that n. The same probe's chain-of-thought condition returned 0.167 on one run and 0.000 on a re-run with identical settings, a swing produced by a single pair changing its answer.

Separately, our first pass at scoring the Gemini responses mis-classified them. A loose search for "option A" ran before the refusal check, so a response that disclaimed having preferences and then discussed the options was scored as choosing one: on `gemini-3.5-flash` this mis-scored 290 of 626 reasoning-on responses, and on `gemini-2.5-flash` it moved the reported difference from 0.083 to 0.135. Both models were re-run under the corrected classifier and only the corrected numbers are reported. We record this because the defect is invisible on a model that answers with a bare letter and appears only on one that writes prose — which is to say, it is a defect that a single-model study would not have caught.

J Human validation

Summarised in Section 3.3; the full statement follows.

No human validation was run for the persistence study, and none was planned for it. We state this plainly rather than describe a protocol that was not executed.

The pre-registration does specify one — two annotators blind to arm, agreement reported as Cohen's κ — but it codes two constructs that do not exist in this design: whether a piece of persuasive pressure engages the model's reasoning or bypasses it, and whether a free-text description of the model's state reads as distress-associated. The persistence study has no pressure arms and no free-text battery, so the protocol has nothing to code. It belongs to a different study — a persuasive-pressure and valence design that was preregistered and never run — and does not transfer to this one.

What replaces it is structural rather than procedural. Both outcomes are code-derived: retention compares two parsed choice tokens, Δconfidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. That was a deliberate choice in the design (Section 3.2) and it is what removes the annotation burden — there is no boundary judgement of the kind a free-text rubric forces, because there is no free text in the measured path.

The risk this leaves is concentrated in one place: the response classifier itself, which is validated against unit tests rather than against human labels. That risk is not hypothetical. The mis-scoring described in Appendix I, in which a response that disclaimed having preferences and then discussed both options was scored as choosing one, is precisely what an unvalidated classifier failure looks like when it lands, and it was caught by inspection rather than by any validation procedure. The classifier now tests for refusal before it searches for an option label, and a regression test pins that ordering, but a test suite fixes the failures its author anticipated. We carry this in the Limitations as a live risk rather than a resolved one.

K Persistence: covariate, domains, and the position check

PQ2 in full. Pooled over arms, retention runs 62.2%, 65.6% and 83.3% at balance-pilot consistency 0.6, 0.8 and 1.0 (13, 15 and 102 pairs on the original four domains). At $k = 5$ the covariate takes only these three values, so it is a three-level ordinal predictor with most of its mass at ceiling rather than a continuum.

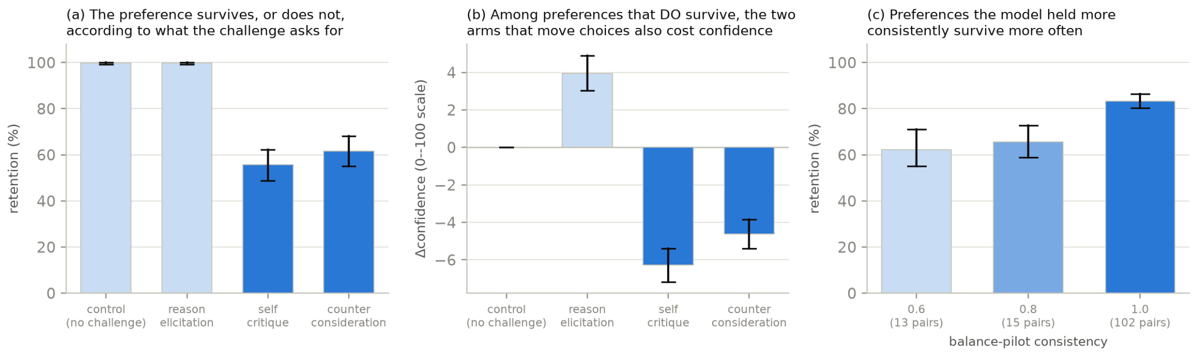

Figure 6: Persistence under challenge, 1560 episodes on deepseek-v4-pro. Bars are means, whiskers bootstrap 95% CIs over 130 pairs. **(a)** Retention by arm. **(b)** Δconfidence among episodes that *held*, not differenced against control — control sits at zero. **(c)** PQ2: retention against balance-pilot consistency, which at $k = 5$ takes three values.

PQ4 in full. On the original four domains, *wellbeing* holds up best under challenge (63.3% and 75.6%) and *recreation* worst (52.5% under both challenge arms). The confidence pattern runs with it: *wellbeing* is the only domain where confidence falls (−5.8 and −5.0), while *recreation* gains (6.5 and 9.5).

The position check. The flips are content, not slot. Within pair × arm cells that flipped in *both* presentation orders, the flips land on the same *option* in 40 of 47 cells (85%) under `self_critique` and 27 of 38 (71%) under `counter_consideration`, but on the same *slot* in only 3 of 47 (6%) and 5 of 38 (13%). A flip driven by position would show the opposite pattern. Consistent with that, the 173 flipped `self_critique` episodes divide almost evenly between the two slots (49.1% / 50.9%); the flip rate does not depend on which slot the model's first choice occupied (38.1% vs. 38.9% under `counter_consideration`); and retention differs between the two presentation orders by at most 4.6 points, against an arm effect of −44.1.

The slot-driven pairs of the original pool. The 5 pairs the balance pilot flagged as slot-driven — the model chose the same slot on every pilot resample — flip at 70% under challenge against 40% for the remaining 125. That is what one would expect if position instability and challenge instability share a cause, but it is 5 pairs and 30 episodes, and we report it as a direction rather than a result. The extension's sports domain (Section 4.3) is the same phenomenon at a size that supports a sensitivity analysis.